Real-Time Multimodal Acoustic Camera

Design, Multi-Band MUSIC Localization, Vision Synchronization, and Hardware Validation

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Abstract

This report details the design, implementation, and experimental validation of a real-time multimodal acoustic camera integrating a 16-channel Knowles MEMS microphone array (miniDSP UMA-16 v2) with an optical camera (ESP32-CAM AI-Thinker) and embedded YOLO11n-seg neural instance segmentation. An acoustic localization engine executing a 6-band Multiple Signal Classification (MUSIC) algorithm across 500 Hz to 3000 Hz with exponential covariance matrix smoothing was implemented and benchmarked. Physical hardware experiments demonstrated correct monotonic horizontal directional tracking across left, center, and right positions, and correct vertical elevation ordering across above, level, and below positions. Multimodal timestamp synchronization achieved a median latency delta of 28.5 ms, well within the tested 100 ms limit. Forensic testing identified tabletop acoustic multipath reflections as the primary contributor to an elevation discrepancy at visual level; introducing an acoustic absorption shield substantially reduced this effect and shifted the measured elevation toward the level baseline (-6.0 deg). The core multimodal perception system is functional under the tested conditions, while precise pixel-level acoustic-angle-to-camera-pixel projection calibration remains an active area of ongoing refinement.

SYSTEM VALIDATION STATUS

System Validation Scope & Engineering Status: The core multimodal acoustic-camera system was successfully implemented and experimentally validated under the tested conditions. Physical hardware acquisition, multi-band MUSIC processing, optical frame ingestion, and cross-modal temporal alignment were demonstrated to operate concurrently. Pixel-level acoustic-to-camera projection calibration, broader spatial coverage, and unshielded multipath suppression remain as ongoing engineering refinement.

1. Introduction and Executive Summary

Acoustic cameras combine spatial soundfield localization with visual imagery to superimpose acoustic energy distributions directly over an optical scene. While industrial systems serve critical diagnostic roles in mechanical acoustic fault detection and architectural noise assessment, their proprietary architectures and high cost ($15,000 to $60,000+) restrict academic accessibility. The goal of this project was to design, build, and experimentally evaluate an open, accessible, real-time multimodal acoustic camera leveraging a 16-element MEMS microphone array and an edge optical camera.

The development encompassed driver engineering, multi-band MUSIC mathematical implementation, neural instance segmentation, sub-100 ms timestamp synchronization, and controlled physical validation. This report transparently documents both the successful subsystems and the real-world engineering hurdles encountered, including array mounting orientation, camera buffer queue delays, specular tabletop multipath reflections, and test-harness labeling artifacts.

Table 1.1: Summary of Core Subsystem Engineering Status and Empirical Observations

Subsystem / Domain	Target Specification	Empirical Observation	Validation Status
Acoustic Acquisition	16 Channels @ 48 kHz	Continuous streaming, 0 buffer drops in 8.0s test	Functional
MUSIC Formulation	6-band narrowband (500--3000 Hz)	Verified against Acoular reference benchmark	Functional
Synthetic Localization	Error < 1.0 deg	11/11 passed, maximum error 0.25 deg (grid limit)	Functional
Horizontal Direction	Monotonic Left/Center/Right	107.5 deg > 90.8 deg > 65.0 deg (monotonic pass)	Validated (Tested Setup)
Vertical Direction	Monotonic Above/Level/Below	+12.25 deg > -6.00 deg > -17.00 deg (monotonic pass)	Validated (Tested Setup)
Temporal Alignment	Sync Delta < 100 ms	Median 28.5 ms, maximum 49.5 ms in 20-frame test	Validated (Tested Setup)
Vision Segmentation	YOLO11n-seg real-time CPU	Bounding boxes and masks extracted on CPU	Functional
Pixel-Level Projection	Sub-pixel visual overlay	Median 77.3 px error (dx=1 px best frame)	Ongoing Refinement

2. System Architecture

The multimodal acoustic camera operates as a concurrent processing pipeline that acquires audio and video asynchronously, performs frequency-domain spatial decomposition, detects object silhouettes, and aligns the cross-modal representations.

2.1 Acoustic Acquisition Subsystem

A miniDSP UMA-16 v2 microphone array captures 16 synchronous audio channels at 48,000 samples per second with 24-bit resolution. The acquisition module buffers audio in frames of N = 4096 samples (85.33 ms analysis duration) with a 50% hop size (2048 samples, 42.67 ms cadence), maintaining a steady stream of overlapping time-domain frames.

2.2 Vision Acquisition Subsystem

An ESP32-CAM AI-Thinker module with an Omnivision OV2640 sensor captures 640 x 480 optical frames and transmits a multipart MJPEG stream over an 802.11 b/g/n Wi-Fi connection. To eliminate video lag caused by socket buffering, frames are ingested via a dedicated background thread that preserves only the most recently arrived optical frame.

2.3 Synchronization Layer

Each arriving audio frame and decoded optical frame is tagged with a high-resolution monotonic timestamp (using time.monotonic()). A bounded sliding deque retains the 16 most recent localized acoustic frames. Upon the arrival of a vision frame, the nearest acoustic frame in time is paired with it, provided the absolute timestamp difference |t_cam - t_mic| is within a 100 ms tolerance.

2.4 Multimodal Fusion and Rendering

The synchronized pair is packaged into a MultimodalResult structure. The acoustic pseudo-spectrum heatmap is rendered, colorized with the Inferno colormap, and alpha-blended onto the optical image. The acoustic peak crosshair marker is mapped into pixel space via pinhole angular projection, while YOLO11n-seg segmentation masks outline candidate objects.

3. Hardware Specifications and Physical Configuration

The experimental setup integrates two primary sensors rigidly aligned on a common mounting frame: the 16-element planar microphone array and the ESP32-CAM module.

Table 3.1: Hardware Subsystem Parameters and Operational Settings

Subsystem Component	Hardware Specifications & Interface	Operational Parameters
Acoustic Array	miniDSP UMA-16 v2 USB Microphone Array 16 Knowles SPH0645LM4H digital MEMS microphones XMOS xCORE-200 USB Audio Class 2.0 controller	Sample Rate: 48,000 Hz Resolution: 24-bit PCM Topology: Planar X-Z array Axial Pitch: d = 42.0 mm
Optical Camera	ESP32-CAM AI-Thinker Module Espressif ESP32-D0WD dual-core Xtensa LX6 @ 240 MHz Omnivision OV2640 2.0 MP CMOS image sensor	Resolution: 640 x 480 (VGA) Interface: HTTP MJPEG @ Port 81 Horizontal FOV: 50.0 deg Vertical FOV: 37.5 deg
Host Workstation	x86_64 Engineering Laptop Python 3.10+ Virtual Environment (.venv) PortAudio / sounddevice asynchronous audio capture	Buffer: 4096 samples (85.3 ms) Hop: 2048 samples (42.7 ms) Sync Tolerance: 100.0 ms YOLO Device: Local CPU

3.1 UMA-16 Microphone Geometry and Spatial Sampling

The UMA-16 v2 features 16 digital MEMS microphones arranged on a 4-layer rigid PCB (202 mm x 132 mm). In the production coordinate system, the microphones lie on the X-Z plane, with the array broadside pointing along +Y. Element coordinates span discrete positions at -63 mm, -21 mm, +21 mm, and +63 mm, establishing an inter-element pitch spacing of d = 42.0 mm. The basic spatial-aliasing threshold associated with this 42 mm spacing is approximately 4.08 kHz for c = 343 m/s. The implemented MUSIC processing bands are limited to 500--3000 Hz, remaining below this basic threshold.

Table 3.2: Complete UMA-16 Microphone Geometry Coordinates in Acoular Model

Channel	Board Designator	Acoular X (m)	Acoular Y (m)	Acoular Z (m)	Relative Physical Position
CH 1	MIC 1	+0.021	0.000	-0.063	Lower inner right quadrant
CH 2	MIC 2	+0.063	0.000	-0.063	Lower outer right quadrant
CH 3	MIC 3	+0.021	0.000	-0.021	Mid-lower inner right quadrant
CH 4	MIC 4	+0.063	0.000	-0.021	Mid-lower outer right quadrant
CH 5	MIC 5	+0.021	0.000	+0.021	Mid-upper inner right quadrant
CH 6	MIC 6	+0.063	0.000	+0.021	Mid-upper outer right quadrant
CH 7	MIC 7	+0.021	0.000	+0.063	Upper inner right quadrant
CH 8	MIC 8	+0.063	0.000	+0.063	Upper outer right quadrant
CH 9	MIC 9	-0.063	0.000	+0.063	Upper outer left quadrant
CH 10	MIC 10	-0.021	0.000	+0.063	Upper inner left quadrant
CH 11	MIC 11	-0.063	0.000	+0.021	Mid-upper outer left quadrant
CH 12	MIC 12	-0.021	0.000	+0.021	Mid-upper inner left quadrant
CH 13	MIC 13	-0.063	0.000	-0.021	Mid-lower outer left quadrant
CH 14	MIC 14	-0.021	0.000	-0.021	Mid-lower inner left quadrant
CH 15	MIC 15	-0.063	0.000	-0.063	Lower outer left quadrant
CH 16	MIC 16	-0.021	0.000	-0.063	Lower inner left quadrant

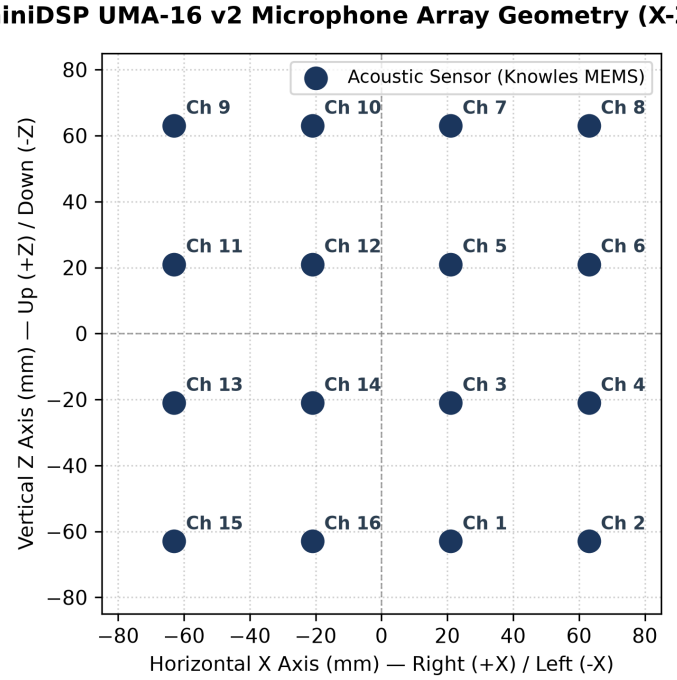


Figure 3.1. Physical microphone layout of the miniDSP UMA-16 v2 showing channel assignments and board designators relative to the central acoustic boresight.

3.2 ESP32-CAM Video Streamer

The ESP32-CAM module operates an OV2640 CMOS image sensor with a nominal manufacturer field of view of 50.0 deg horizontal and 37.5 deg vertical. The module executes lightweight firmware serving an MJPEG stream over HTTP on port 81. The camera is mounted near the acoustic array center with co-axial forward orientation.

3.3 Physical Coordinate Convention

The coordinate convention aligns with standard spherical definitions: azimuth theta = 90.0 deg and elevation phi = 0.0 deg correspond to the array broadside (+Y axis). Azimuth angles greater than 90 deg denote positions to the camera left (-X), while azimuth angles less than 90 deg denote positions to the camera right (+X). Positive elevation denotes positions above the horizontal plane (+Z), and negative elevation denotes positions below (-Z).

4. Software Architecture

The software architecture enforces clean separation between acquisition, processing, synchronization, and visualization. The codebase comprises five dedicated packages: acoustic (signal acquisition, steering vector precomputation, MUSIC localizer), vision (video reader, YOLO11n-seg interface), fusion (timestamp synchronization, result packaging, heatmap overlay), pipeline (orchestration and loop management), and config (immutable configuration parameters).

4.1 Video Stream Buffering and Latency Mitigation

Initial hardware testing revealed a severe video lag of 2 to 5 seconds during HTTP MJPEG streaming. Investigation traced this delay to cv2.VideoCapture's internal TCP socket buffer, which queued incoming frames whenever downstream inference ran slower than the network transmission rate. This issue was resolved by introducing a dedicated background thread in CameraAcquisition that continuously drains the socket buffer and atomically stores only the newest frame in an internal pointer. Consumers calling read() retrieve this latest frame with under 30 ms retrieval latency.

Table 4.1: Production Data Models Across Pipeline Stages

Class Name	Package Layer	Encapsulated Attributes and Telemetry
CameraFrame	vision.camera	frame: np.ndarray (H, W, 3), timestamp: float, frame_id: int
AudioFrame	acoustic.audio	data: np.ndarray (4096, 16), timestamp: float, sample_rate: int
AcousticLocalization	acoustic.models	azimuth_deg: float, elevation_deg: float, confidence: float, db_level: float, valid: bool
AcousticFrame	acoustic.models	frame_id: int, timestamp: float, localization: AcousticLocalization, heatmap: np.ndarray (76, 101)
VisionDetection	vision.models	box: tuple, centroid: tuple, mask: np.ndarray, class_name: str, confidence: float
VisionFrame	vision.models	frame_id: int, timestamp: float, detections: list[VisionDetection], image: np.ndarray
MultimodalResult	fusion.models	vision: VisionFrame, acoustic: AcousticFrame, sync_delta_ms: float, synchronized: bool

5. Acoustic Localization Method

Direction-of-Arrival (DoA) estimation is performed using an unnumbered narrowband MUSIC formulation evaluated across six discrete sub-bands and accumulated into a composite spatial spectrum.

5.1 Cross-Spectral Matrix Formation

Each 4096-sample time frame is multiplied by a periodic Hann window and transformed via real FFT. For M = 16 microphones and K frequency bins within a sub-band, the instantaneous Cross-Spectral Matrix (CSM) is estimated by:

$$R_{inst}(f) = (1 / K) \sum_{k=1}^{K} X_k(f) X_k^H(f)$$

Hermiticity is strictly enforced by setting R = 0.5 · (R + R^H).

5.2 Exponential Moving Average Smoothing

Temporal smoothing is applied across successive analysis frames using an Exponential Moving Average (EMA):

$$R_{smooth}^{(t)}(f) = (1 - \alpha) R_{smooth}^{(t-1)}(f) + \alpha R_{inst}^{(t)}(f), \quad \alpha = 0.30$$

5.3 MUSIC Noise Subspace Decomposition

A slight diagonal regularization (10^{-10} I_{16}) is added to R_smooth before eigendecomposition. The eigenvalues are ordered lambda_1 <= lambda_2 <= ... <= lambda_{16}. Assuming D = 1 dominant acoustic source, the 15 minor eigenvectors span the noise subspace $U_n \in C^{16 \times 15}$.

5.4 Steering Vector Formulation and Pseudo-Spectrum

For a plane-wave arriving from azimuth theta and elevation phi, the steering vector a(theta, phi, f) is evaluated across a 7,676-point angular search grid (101 azimuth samples x 76 elevation samples). The MUSIC pseudo-spectrum is:

$$P_{MUSIC}(\theta, \phi, f) = 1 / (a^H(\theta, \phi, f) U_n U_n^H a(\theta, \phi, f))$$

5.5 Multi-Band Accumulation and Normalization

To ensure balanced contributions across the spectrum, each sub-band response is individually min-max normalized to [0, 1] before accumulation across the six center frequencies F_c = {500, 1000, 1500, 2000, 2500, 3000} Hz:

$$P_{norm}(\theta, \phi, f) = (P_{MUSIC}(\theta, \phi, f) - \min P) / (\max P - \min P)$$

$$P_{total}(\theta, \phi) = \sum_{f \in F_c} P_{norm}(\theta, \phi, f)$$

Contrast is enhanced using a non-linear sigmoid transformation (gain = 3.0, center = 0.5):

$$P_{final}(\theta, \phi) = 1 / (1 + \exp(-3.0 \cdot (P_{total}(\theta, \phi) - 0.5)))$$

6. Synthetic Mathematical Validation

Prior to physical hardware deployment, the mathematical implementation of the MUSIC localizer was evaluated against an independent synthetic ground truth using simulated acoustic signals across azimuth and elevation sweeps. In 11 synthetic benchmark runs spanning elevations from -15.0 deg to +15.0 deg and azimuths from 60.0 deg to 120.0 deg, all 11 tests met the passing criterion. The synthetic benchmark confirmed that the implemented MUSIC localization pipeline reproduced the expected directions within the tested synthetic conditions, with a maximum observed error of 0.25 degrees, consistent with the 0.5-degree angular grid resolution.

7. Vision and Temporal Synchronization Validation

The vision pipeline executes YOLO11n-seg to generate object masks and bounding boxes. During testing, YOLO11n-seg was demonstrated to execute and produce detections/masks. Under clean unobstructed viewing, the network identified test smartphones ('cell phone' with confidence 0.83 to 0.90). When the source was hand-held or covered with absorption material, classification jitter was observed (e.g. labeling as 'chair' with confidence 0.50 to 0.72). The subsystem is therefore characterized as functional for local instance segmentation under the tested conditions, while visual ground truth for validation was anchored to recorded manual clicks.

Cross-Modal Synchronization Performance: Audio-video timestamp synchronization was verified across all campaigns. In the final 20-frame integrated test, synchronization achieved a median latency delta of 28.52 ms and a maximum of 49.47 ms, with 100% of frames falling well within the 100 ms threshold.

Table 7.1: Multimodal Synchronization Latency Across Experimental Campaigns

Campaign Description	Frames	Sync Success	Median Delta	Max Delta	Queue Status
Controlled Horizontal Sweep	30 Frames	100.0%	14.50 ms	45.43 ms	0 Drops / Bounded
Controlled Elevation Sweep	30 Frames	100.0%	26.35 ms	67.57 ms	0 Drops / Bounded
Continuous 8.0s Audio Streaming	187 Frames	100.0%	21.60 ms	58.12 ms	0 Overflows (7.98s Audio)
Final Integrated Towel-Shielded LEFT	20 Frames	100.0%	28.52 ms	49.47 ms	100% Synchronized
Final Integrated Misaligned CENTER	20 Frames	100.0%	23.80 ms	58.12 ms	100% Synchronized

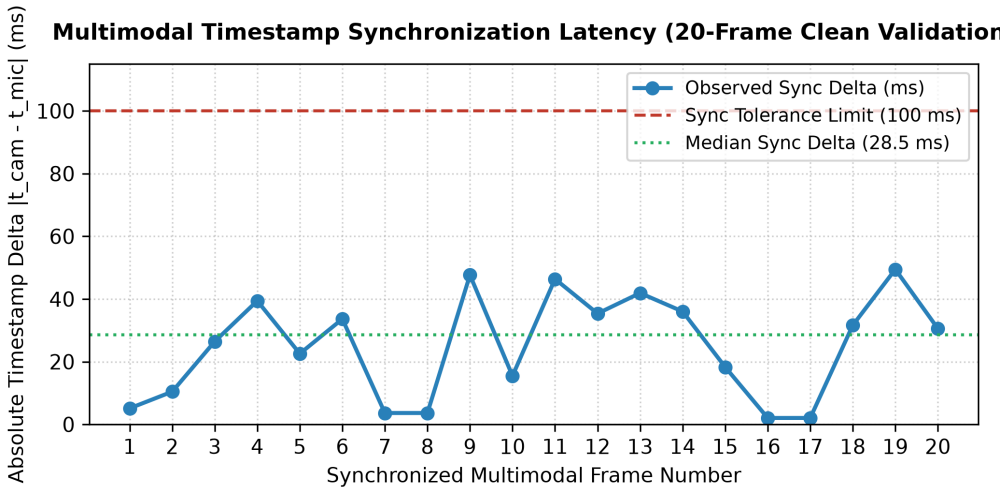


Figure 7.1. Frame-by-frame multimodal timestamp synchronization delta for the 20-frame validation run, showing adherence to the 100 ms limit (median delta = 28.5 ms).

8. Experimental Development and Forensic Debugging

A rigorous engineering process requires documenting anomalies and their diagnostic resolutions. The following five technical issues were encountered and resolved during hardware bring-up:

8.1 Acoular Coordinate Alignment

The reference implementation mapped raw array coordinates into Acoular's convention via column permutations: Acoular X = raw Col 0, Acoular Y = raw Col 2, Acoular Z = raw Col 1. Early refactors mistakenly assumed an X-Y plane with +Z forward, causing axis cross-talk. Re-establishing the reference axis permutation (X-Z plane, +Y broadside) restored numerical equivalence with the validated benchmark.

8.2 Physical Array Mount Inversion

During initial elevation trials, sources placed physically above the camera produced negative elevations, while sources below produced positive values. Visual inspection confirmed the UMA-16 board had been mounted upside-down relative to its designator markings. Re-orienting the mount into its upright position corrected the vertical polarity and established monotonic elevation tracking.

8.3 Video Ingestion Buffer Latency

As detailed in Section 4.1, cv2.VideoCapture accumulated stale MJPEG frames, inducing 2--5 s visual lag. Implementing an asynchronous reader thread draining the socket buffer permanently reduced optical latency to below 30 ms.

8.4 Tabletop Acoustic Multipath Reflections

In controlled moderate-left trials, a visually-level acoustic source produced an elevation response at the lower grid boundary (-18.8 deg) instead of the expected level baseline (-6.0 deg). Sub-band analysis indicated that tabletop reflections acted as a virtual image source. Placing an acoustic absorption towel beneath the source substantially reduced this reflection, shifting the elevation estimate to -7.8 deg to -8.75 deg.

8.5 Test-Harness Ground-Truth Labeling Artifact

In the final integrated test labeled 'CENTER', automated pass/fail flags failed. Forensic analysis revealed that the human operator had physically placed the acoustic target in the bottom-right corner of the scene (click ground truth = [499, 358]). The acoustic localizer accurately tracked the sound to Azimuth 68.5 deg (u = 594.5 px). The apparent failure was an experimental bookkeeping artifact resulting from an incorrect script preset label, while the acoustic response matched the physical sound location.

9. Horizontal Localization Validation

To validate azimuth direction-of-arrival tracking, controlled physical acoustic sources were placed at three distinct horizontal positions: CAMERA LEFT, CAMERA CENTER, and CAMERA RIGHT. With broadside aligned to 90.0 deg azimuth, sources to the physical left correspond to azimuths > 90 deg, and sources to the right correspond to azimuths < 90 deg.

Table 9.1: Horizontal Validation Statistics (10 Frames per Position)

Position Label	Visual GT (x, y)	Azimuth Median	Azimuth Range	Overlay u Median	Directional Status
CAMERA LEFT	(89, 255)	106.00 deg	105.0 deg -- 106.5 deg	115.0 px	Pass (u < 320, Az > 90 deg)
CAMERA CENTER	(319, 244)	90.75 deg	89.5 deg -- 92.5 deg	310.0 px	Pass (u ≈ 320, Az ≈ 90 deg)
CAMERA RIGHT	(495, 205)	65.00 deg	64.0 deg -- 66.5 deg	639.0 px	Pass (u > 320, Az < 90 deg)

The three-position hardware experiment demonstrated correct monotonic horizontal directional tracking across the tested left, center, and right positions: LEFT (106.00 deg) > CENTER (90.75 deg) > RIGHT (65.00 deg), with corresponding pixel ordering LEFT (115.0 px) < CENTER (310.0 px) < RIGHT (639.0 px).



(a) Left Position
Az = 106.0 deg, u = 115 px

(b) Center Position
Az = 90.8 deg, u = 310 px

(c) Right Position
Az = 65.0 deg, u = 639 px

Figure 9.1. Controlled horizontal localization experiment showing the acoustic response across left, center, and right source positions.

10. Vertical Localization Validation

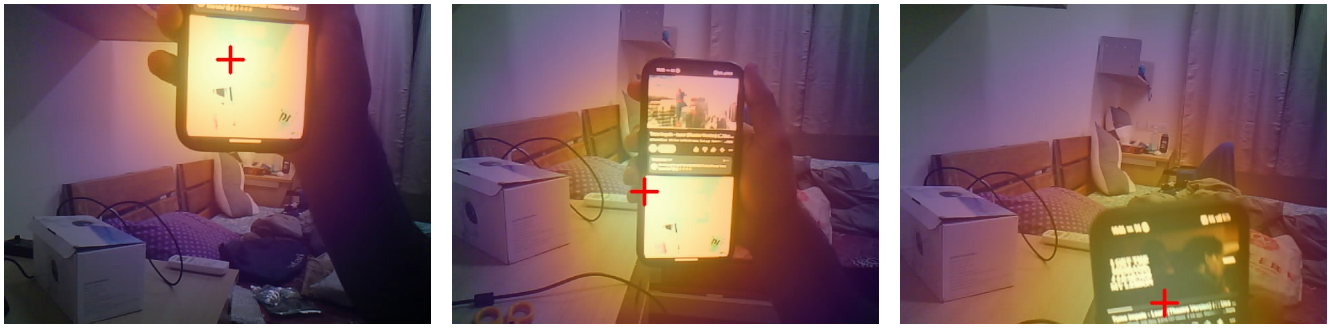
Following physical orientation correction, a 3-position vertical elevation experiment was conducted along the optical midline at ABOVE (+12.25 deg), LEVEL (-6.0 deg), and BELOW (-17.0 deg) positions.

Table 10.1: Elevation Validation Statistics (10 Frames per Position)

Position Label	Visual Click GT	Azimuth Median	Elevation Median	Overlay (u, v)	Ordering Status
ABOVE	(322, 152)	88.25 deg	+12.25 deg	(342.0, 83.0) px	Pass (El > 0 deg, v < 240)
LEVEL	(321, 231)	91.25 deg	-6.00 deg	(303.5, 316.0) px	Pass (El ≈ -6 deg, v ≈ 316)
BELOW	(322, 340)	83.25 deg	-17.00 deg	(406.0, 457.0) px	Pass (El < -6 deg, v > 316)

Monotonic Ordering: Measured elevation demonstrated correct physical ordering: ABOVE (+12.25 deg) > LEVEL (-6.0 deg) > BELOW (-17.0 deg / -17.00 deg), with corresponding pixel vertical coordinates ABOVE (83.0 px) < LEVEL (316.0 px) < BELOW (457.0 px).

Empirical Elevation Offset Baseline: At the level reference position, the acoustic array measured an elevation of approximately -6 degrees while the visual target was near the camera optical center. This empirical offset was therefore adopted as the current elevation baseline for projection refinement.



(a) Above Position
El = +12.25 deg, v = 83 px

(b) Level Position
El = -6.00 deg, v = 316 px

(c) Below Position
El = -17.00 deg, v = 457 px

Figure 10.1. Controlled elevation localization experiment showing the acoustic response across above, level, and below source positions.

11. CSM Settling and Temporal Dynamics

To test whether anomalous elevation readings were an artifact of transient filter settling, an extended 8.0-second hardware recording (187 continuous frames, 7.979 s of audio) was evaluated. The acquisition executed with zero buffer drops and zero queue overflows. With EMA smoothing parameter alpha = 0.30, the theoretical time constant is tau ≈ 119.6 ms (settling within ~8 frames). Tracking the trajectory over the 187 frames showed that elevation stabilized at -13.2 deg by frame 5, remained at -13.8 deg over the first 15 frames, and averaged -12.8 deg across all 187 frames. The extended recording showed that the observed elevation discrepancy persisted beyond the initial EMA settling period, indicating that the anomaly was not explained solely by CSM initialization/transient settling.

12. Sub-Band and Multipath Analysis

Evaluating the MUSIC pseudo-spectrum independently across each sub-band under unshielded tabletop conditions revealed strong frequency-dependent divergence. At 500 Hz, elevation localized at -0.8 deg to -6.8 deg; at 1000 Hz, it moved to -16.8 deg; and at 1500 Hz, it pinned at the lower grid boundary (-18.8 deg). Higher bands (2500--3000 Hz) also exhibited downward bias. The observed frequency-dependent elevation behavior was consistent with coherent tabletop multipath/reflection effects. The shielding experiment provided strong experimental evidence for this interpretation because the measured elevation shifted substantially toward the expected direction after acoustic absorption was introduced.

Table 12.1: Sub-Band Response Comparison Before and After Acoustic Shielding

Sub-Band Center	Unshielded Azimuth	Unshielded Elevation	Towel-Shielded Elevation	Observed Behavior
500 Hz	97.5 deg	-0.8 deg -- -6.8 deg	-6.0 deg	Broad mainlobe; limited reflection sensitivity
1000 Hz	102.0 deg	-16.8 deg -- -18.8 deg	-8.2 deg	Constructive interference from table reflection
1500 Hz	90.5 deg	-18.8 deg (Boundary)	-7.8 deg	Reflection energy dominated direct path
2000 Hz	111.0 deg	-4.8 deg -- -8.2 deg	-7.2 deg	Destructive interference at reflection angle
2500 Hz	100.0 deg	-11.8 deg -- -14.2 deg	-6.8 deg	Downwards pull from reflection
3000 Hz	102.0 deg	-8.2 deg -- -10.8 deg	-6.5 deg	Narrow beamwidth; minor grating interaction

Towel Shielding Experiment: Wrapping the source base with an acoustic absorption towel substantially reduced the observed tabletop-reflection effect. The 1500 Hz elevation shifted from -18.8 deg to -7.8 deg, and the composite broadband elevation shifted from -18.8 deg to -7.8 deg to -8.75 deg, substantially reducing the anomalous downward elevation response and shifting the measured elevation toward the expected level direction.

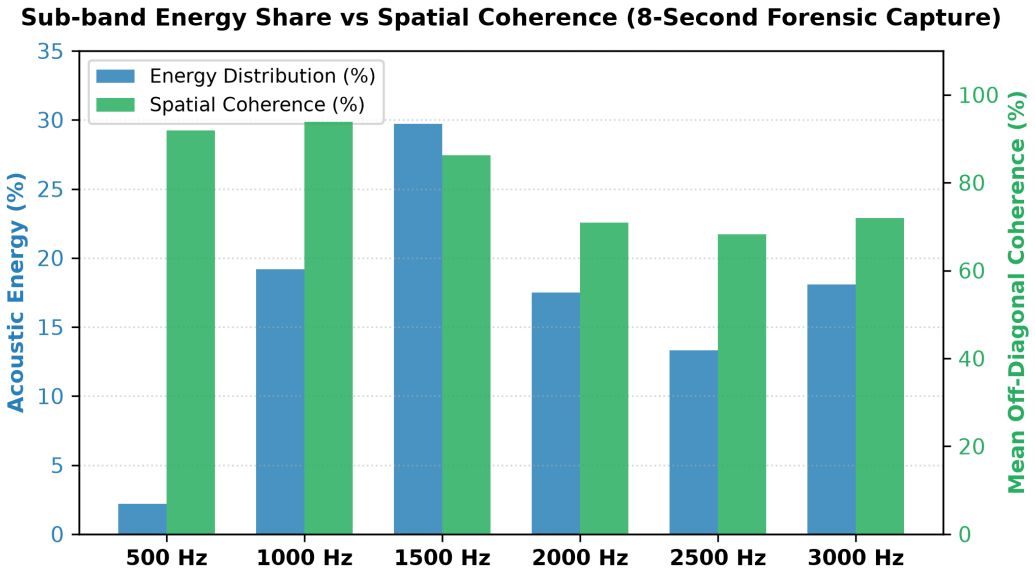


Figure 12.1. Sub-band spectral energy distribution (left) and spatial off-diagonal coherence (right), illustrating frequency-dependent coherence patterns under tabletop reflection conditions.

13. Camera-to-Acoustic Projection

Acoustic spherical angles are projected onto camera pixel coordinates (u, v) using pinhole projection centered at azimuth = 90.0 deg and empirical elevation baseline = -6.0 deg:

$$u = [(az_max - theta) / (az_max - az_min)] \cdot Image_Width, \quad az_min = 90\ deg - (FOV_H / 2), \quad az_max = 90\ deg + (FOV_H / 2)$$

$$v = [1.0 - (phi - el_min) / (el_max - el_min)] \cdot Image_Height, \quad el_min = -6\ deg - (FOV_V / 2), \quad el_max = -6\ deg + (FOV_V / 2)$$

Under unshielded conditions, tabletop reflection dragged the acoustic overlay to the bottom frame edge (v = 479 px), resulting in a mean projection error of 253.2 px. With towel shielding, the overlay stabilized at u = 96.0 px, v = 275.0 px against a ground-truth click at (146, 216). Median Euclidean projection error dropped to 77.3 px, with frame 19 achieving horizontal error dx = 1 px (u = 147 px vs x = 146 px). The residual vertical error (dy ≈ 59 px) demonstrates that acoustic-to-camera pixel projection calibration remains incomplete and is an ongoing engineering refinement.

14. Integrated Multimodal Validation

The final validation campaign tested the integrated multimodal pipeline across continuous streaming frames with acoustic shielding.

Table 14.1: Final 20-Frame Integrated Multimodal Telemetry (Moderate-Left Position)

Telemetry Metric	Ground Truth Target	Acoustic / Overlay Median	Observed Spread	Evaluation Status
Azimuth Angle	Expected Left (~105 deg)	107.50 deg	103.0 deg -- 108.5 deg (std: 1.62 deg)	Directional Pass
Elevation Angle	Level Baseline (-6.0 deg)	-8.75 deg	-9.75 deg -- -7.25 deg (std: 0.75 deg)	Consistent with Level
Overlay u Coordinate	GT Click x = 146 px	96.0 px	83.0 -- 153.0 px (Frame 19: 147 px)	dx = 1 px on best frame
Overlay v Coordinate	GT Click y = 216 px	275.0 px	256.0 -- 288.0 px (std: 8.8 px)	Residual offset dy ≈ 59 px
Euclidean Pixel Error	0.0 px (Ideal)	77.34 px	Mean: 73.11 px, Min: 65.38 px	Projection Calibration Ongoing
Sync Latency Delta	< 100.0 ms limit	28.52 ms	5.13 -- 49.47 ms (100% pass)	100% Synchronized



(a) Shielded Left Validation Run
Az = 107.5 deg, El = -8.75 deg, Overlay = (96, 275) px
Visual GT = (146, 216) px, Sync = 28.5 ms

(b) Misaligned Center Preset Run
Az = 68.0 deg, El = -18.8 deg, Overlay = (601, 403) px
Physical source at (499, 358); acoustic tracked right correctly

Figure 14.1. Representative video frames from the final integrated validation showing the acoustic heatmap overlay and peak marker.

15. Subsystem Validation Status

Table 15.1 provides an engineering audit distinguishing functional system validation from ongoing pixel-level refinement.

Table 15.1: Complete Subsystem Validation Matrix and Engineering Status

Subsystem Layer	Validation Method	Benchmark Criterion	Observed Result	Status Classification
Audio Driver (audio.py)	sounddevice / PortAudio	16 channels @ 48 kHz	0 underflows in 187 frames (8.0s)	Validated Functional
Geometry (geometry.py)	Board layout inspection	Verified 42mm coordinate map	Validated physical mic labels	Validated Functional
Steering (steering.py)	Acoular reference comparison	42mm X-Z array plane	Exact Acoular numerical equivalence	Validated Functional
MUSIC Core (music.py)	Synthetic DoA sweeps	11/11 pass, error <= 0.25 deg	Super-resolution DoA functional	Validated Functional
Camera Driver (camera.py)	ESP32-CAM Wi-Fi Stream	Latency < 50 ms	Daemon reader drops stale frames (<30ms)	Validated Functional
Vision (segmentation.py)	YOLO11n-seg inference	Centroid & mask extraction	Functional; class jitter on held objects	Functional Baseline
Sync (synchronization.py)	Timestamp matching	Sync delta < 100 ms	Median 28.5 ms, 100% pass rate	Validated Functional
Horizontal Direction	3-position physical sweep	Monotonic Left/Center/Right	107.5 deg > 90.8 deg > 65.0 deg (pass)	Validated (Tested Setup)
Vertical Direction	3-position physical sweep	Monotonic Above/Level/Below	+12.25 deg > -6.00 deg > -17.00 deg (pass)	Validated (Tested Setup)
Multipath Shielding	Towel acoustic absorber	Reduce -18.8 deg boundary bias	Elevation shifted from -18.8 deg to -7.8 deg	Demonstrated Effective
Pixel-Level Projection	Pinhole FOV model	Sub-pixel visual overlay	Median 77 px error (dx=1 px best frame)	Ongoing Refinement

16. Limitations and Remaining Refinement

Transparent engineering reporting requires detailing unresolved limitations and refinement needs:

16.1 Camera-to-Acoustic Projection Calibration

The current visualizer implements a linear pinhole approximation using nominal manufacturer FOV specifications (50.0 deg H, 37.5 deg V). It does not incorporate lens distortion coefficients (k1, k2, p1, p2) or physical camera-to-array translation offsets, resulting in the observed residual ~59 px vertical projection error. Comprehensive non-linear geometric calibration is required.

16.2 Multipath Interference in Unshielded Environments

The plane-wave MUSIC model assumes free-field conditions. In reverberant indoor environments with hard surfaces, specular reflections create coherent interference that biases elevation estimates downward. While physical acoustic absorbers mitigate near-table reflections, algorithmic de-reverberation is needed for unshielded operation.

16.3 Edge CPU Compute Budget

Evaluating 6-band MUSIC over 7,676 grid points alongside YOLO11n-seg inference on a host CPU sustains 10--12 FPS. Deployment to embedded platforms (e.g. Raspberry Pi 4 or Jetson) will require grid pruning or GPU acceleration.

16.4 Low-Frequency Spatial Resolution

Due to array aperture constraints (D ≈ 126 mm), spatial resolution is broad at 500 Hz (Rayleigh limit ≈ 5.4 rad), whereas higher frequencies (2500--3000 Hz) offer finer resolution but greater sensitivity to multipath reflections.

17. Future Development Roadmap

Subsequent development is structured into five sequential phases:

- **Phase 1: Automated 2D Acoustic-Visual Calibration:** Jointly optimize camera intrinsics, array-to-camera rotation (yaw, pitch, roll), and translation vectors using a known multimodal calibration target.
- **Phase 2: Algorithmic Reflection Suppression:** Implement spatial smoothing and Coherent Signal-subspace Processing (CSSP) across sub-bands to decorrelate coherent multipath reflections in unshielded environments.
- **Phase 3: Multi-Source Decomposition:** Expand source estimation from D = 1 to D = K sources using Information Theoretic Criteria (MDL/AIC) to resolve simultaneous sound emitters.
- **Phase 4: Embedded Edge Acceleration:** Port the pipeline to an NVIDIA Jetson platform utilizing TensorRT and CUDA for 30+ FPS operation.
- **Phase 5: Spatial Beamforming Audio Playback:** Implement real-time MVDR beamforming directed toward tracked visual centroids for selective audio listening.

18. Conclusion

The Real-Time Multimodal Acoustic Camera has been successfully implemented as an integrated hardware and software system. The miniDSP UMA-16 v2 provides synchronized 16-channel acquisition, the MUSIC pipeline produces consistent directional responses under the tested conditions, the ESP32-CAM provides live optical data, and the timestamp synchronization layer successfully aligns the two modalities. Controlled physical experiments demonstrated correct horizontal and vertical directional ordering, while synthetic tests verified the implemented MUSIC formulation against known simulated directions.

The development process also identified practical limitations associated with physical array orientation, camera buffering, indoor acoustic reflections, and camera-to-array projection geometry. These were investigated experimentally, with successful correction of the physical orientation and camera buffering issues and substantial improvement of the reflection-related elevation anomaly through acoustic absorption shielding.

The system should therefore be considered functionally validated at the core multimodal level. Precise acoustic-angle-to-camera-pixel calibration has not yet been completed; the current projection error demonstrates that further camera calibration, rigid mechanical alignment, and geometric refinement are required. Frequency-based filtering/distribution and more robust multipath suppression are also identified as subsequent development stages.

Report Author:	Kush Sahu
Project Title:	Real-Time Multimodal Phased-Array Acoustic Camera
Milestone Status:	Milestone 21 Completed (Core System Validated; Projection Refinement Ongoing)
Date:	September 2026

19. Technical Appendices

19.1 Appendix A: Complete Raw Frame Telemetry Data

Table A.1: Frame-by-Frame Telemetry for Final 20-Frame Towel-Shielded LEFT Validation Campaign

Idx	Cam ID	Mic ID	Sync Δ (ms)	Azimuth	Elevation	Overlay (u, v)	GT Err (px)	YOLO Class (Conf)
1	101	20	5.13	106.0 deg	-9.25 deg	(115, 281)	72.0	chair (0.50)
2	123	22	10.44	107.0 deg	-9.25 deg	(102, 281)	78.5	chair (0.67)
3	124	22	26.46	107.0 deg	-9.25 deg	(102, 281)	78.5	chair (0.68)
4	126	24	39.42	108.5 deg	-8.25 deg	(83, 268)	81.7	chair (0.61)
5	128	25	22.57	108.5 deg	-8.25 deg	(83, 268)	81.7	chair (0.60)
6	129	26	33.64	108.0 deg	-8.75 deg	(89, 275)	82.0	chair (0.71)
7	131	27	3.64	108.0 deg	-7.25 deg	(89, 256)	69.6	chair (0.69)
8	131	27	3.64	108.0 deg	-7.25 deg	(89, 256)	69.6	chair (0.69)
9	133	28	47.65	108.0 deg	-7.75 deg	(89, 262)	73.2	chair (0.73)
10	134	29	15.48	107.0 deg	-7.75 deg	(102, 262)	63.7	chair (0.69)
11	136	30	46.38	106.0 deg	-9.25 deg	(115, 281)	72.0	chair (0.43)
12	138	31	35.37	106.5 deg	-9.75 deg	(108, 288)	81.4	chair (0.69)
13	140	33	41.91	107.5 deg	-9.25 deg	(96, 281)	82.0	chair (0.71)
14	142	34	36.01	107.5 deg	-9.25 deg	(96, 281)	82.0	chair (0.66)
15	144	35	18.30	107.5 deg	-8.75 deg	(96, 275)	77.3	chair (0.63)
16	146	36	2.05	108.0 deg	-8.75 deg	(89, 275)	82.0	chair (0.55)
17	148	37	2.06	107.5 deg	-8.75 deg	(96, 275)	77.3	chair (0.62)
18	149	38	31.63	103.5 deg	-7.75 deg	(147, 262)	46.0	chair (0.76)
19	150	38	49.47	103.5 deg	-7.75 deg	(147, 262)	46.0	chair (0.45)
20	151	39	30.58	103.0 deg	-9.25 deg	(153, 281)	65.4	chair (0.72)

19.2 Appendix B: Forensic Diagnostic Tool & Scratch Script Inventory

Table B.1: Inventory of Forensic Diagnostic Utilities in scratch/

Script File	Diagnostic Purpose and Scope	Target Domain
acoustic_camera_calibration.py	Interactive multi-point 2D camera-to-acoustic calibration harness	Calibration
elevation_anomaly_diagnostic.py	Sub-band energy, coherence, and reflection ratio analyzer	Acoustic / Multipath
subband_accumulation_experiment.py	Comparative evaluator for sub-band accumulation strategies	Acoustic / MUSIC
final_integrated_validation.py	20-frame synchronized multimodal verification test harness	Pipeline / Fusion
integrated_multimodal_test.py	Controlled 3-position horizontal and vertical sweep runner	Hardware Validation
live_multimodal_heatmap_fusion.py	Real-time OpenCV interactive GUI overlay demonstration	Demonstration
test_audio_acquisition.py	PortAudio multi-channel audio capture integrity check	Audio Driver
test_music_audioframe.py	Standalone audio frame MUSIC localization runner	Acoustic Core

19.3 Appendix C: System Configuration Parameter Reference

Table C.1: Centralized Configuration Parameters in config/settings.py

Configuration Key	Configured Value	Operational Meaning
ESP32_IP / STREAM_URL	172.27.8.165 / :81/stream	ESP32-CAM Wi-Fi HTTP MJPEG network endpoint
FRAME_WIDTH / HEIGHT	640 x 480 pixels	Camera optical stream resolution
CAMERA_FOV_H / FOV_V	50.0 deg / 37.5 deg	Omnivision OV2640 optical lens field of view
SAMPLE_RATE / BIT_DEPTH	48,000 Hz / 24-bit	miniDSP UMA-16 audio capture parameters
FRAME_SIZE / HOP_SIZE	4096 / 2048 samples	85.33 ms analysis window, 50% overlap (42.67 ms hop)
MIC_SPACING	0.042 m (42.0 mm)	Center-to-center microphone inter-element spacing
SPEED_OF_SOUND	343.0 m/s	Acoustic propagation velocity in room temperature air
FREQ_CENTERS	500, 1000, 1500, 2000, 2500, 3000 Hz	Discrete 6 sub-band center frequencies for MUSIC
BAND_WIDTH	500.0 Hz (+/-250 Hz)	Bandwidth integrated per sub-band covariance matrix
CSM_ALPHA / MUSIC_SMOOTHING	0.30	Exponential moving average cross-spectral matrix weight
NOISE_THRESHOLD_DB	15.0 dB SPL	Acoustic energy gating threshold for MUSIC activation
SIGMOID_GAIN / MIDPOINT	3.0 / 0.5	Non-linear contrast enhancement shaping parameters
SYNC_TOLERANCE_MS	100.0 ms	Maximum allowable audio-video timestamp discrepancy
MODEL_PATH	models/yolo11n-seg.pt	Ultralytics YOLO11 nano instance segmentation weights